DuQuantum Pasqal Challenge

Quantum Optimization with Pasqal neutral atoms

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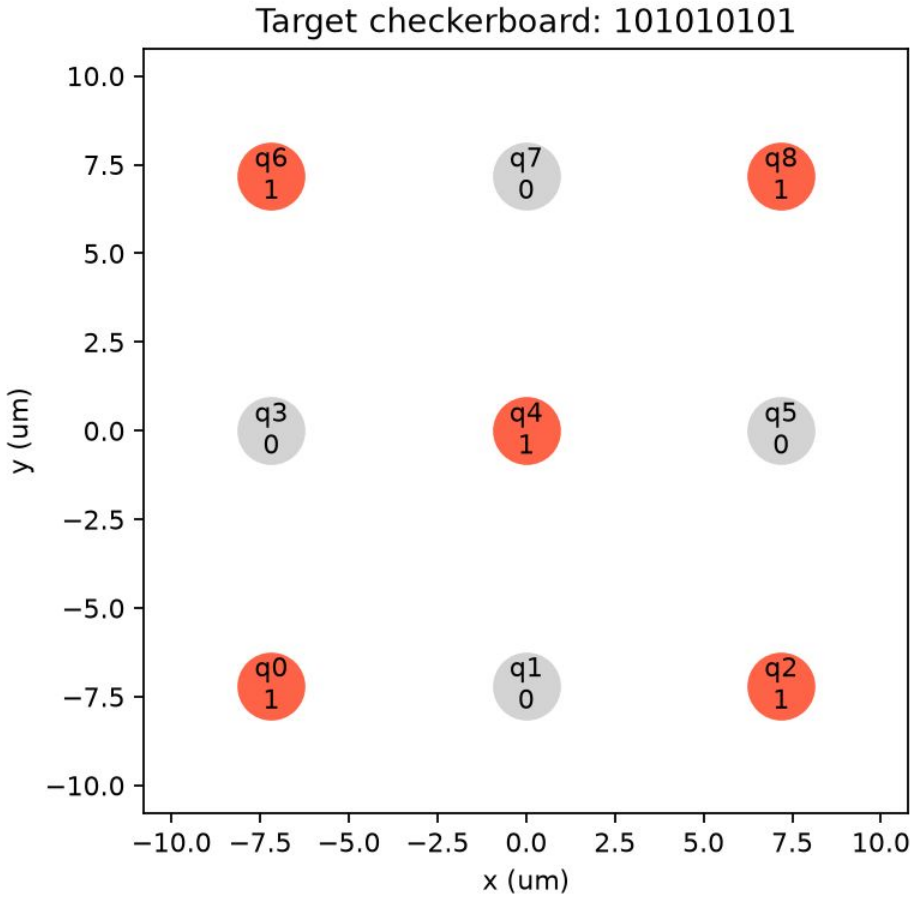
Pasqal and neutral-atom optimization

Analog quantum processors from neutral atoms held in optical traps.

Lasers drive atoms between a ground state and an excited Rydberg state.

Rydberg interactions set pairwise costs. Lasers set selection costs and drive state preparation.

Pulser specifies the register and pulses, then samples binary outcomes.



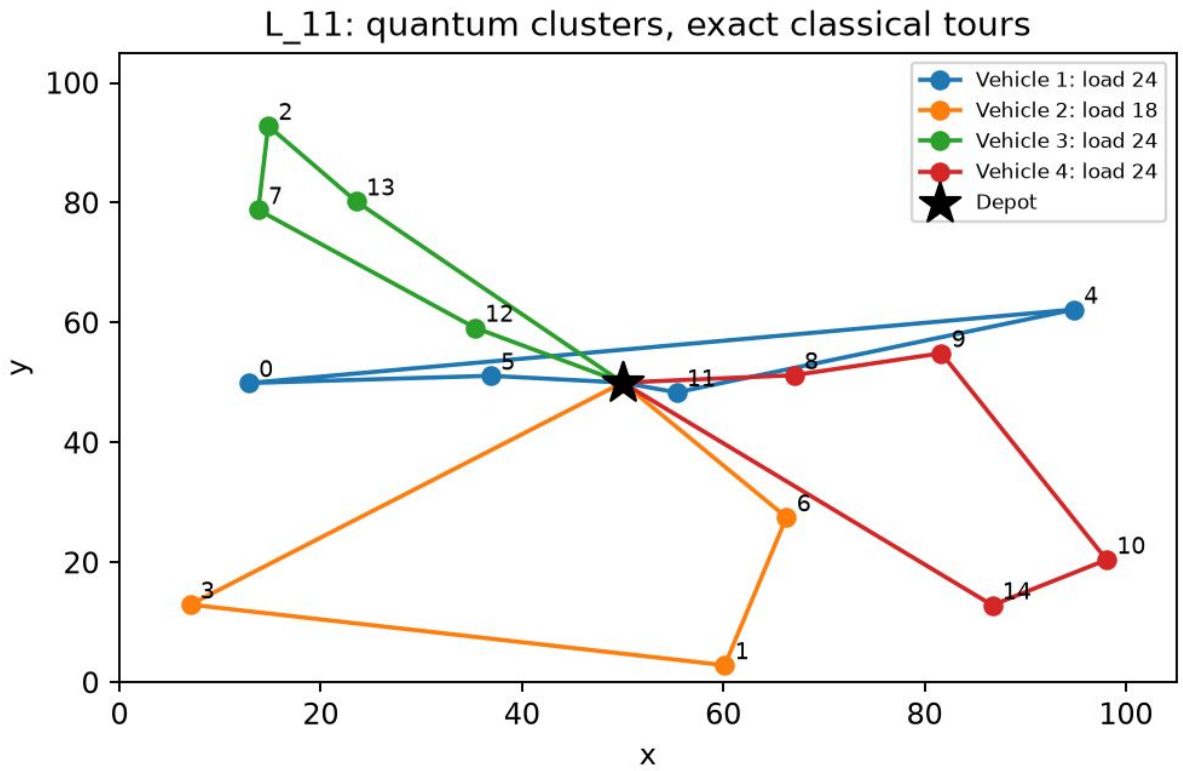
Saved nine-atom tutorial register; nearest-neighbor spacing 7.19 μm.

The capacitated vehicle routing problem

Tier	Customers	Vehicles	Capacity
S	4	2	7
M	10	3	20
L	15	4	30

Minimize total depot-to-depot distance.

Serve every customer once.
Respect each vehicle’s capacity.
Allow unused vehicles.



Saved L_{11} example: quantum clustering with exact classical tours

QUBO formulation and capacity slack

Binary decisions become an energy

$$E(\mathbf{b}) = c_0 + \sum_i h_i b_i + \sum_{i<j} J_{ij} b_i b_j, \qquad b_i \in \{0, 1\}$$

Linear terms score selections. Pair terms score joint choices.

Slack represents unused vehicle capacity

$$3x_1 + 2x_2 + s = 7, \qquad s = u_0 + 2u_1 + 4u_2$$

Load 5 + unused capacity 2 = capacity 7

The exact slack model is a reference. Our executed sampler checks capacity classically.

The implemented hybrid solver

Propose the next group

$$E_{\mathrm{cl}}/\kappa = \lambda m^2 - \mu m + \sum_{i<j} w_{ij} b_i b_j$$

$$m = \sum_i b_i, \qquad \mu = 2\lambda \frac{|R|}{L}$$

At most 8 candidate customers per call.
Group size is centered near |R|/L.
Classical checks reject overloaded proposals.

λ = 0.15; κ > 0 sets the energy scale.

Order each group's stops

$$E_{\mathrm{edge}} = \sum_e \tilde{c}_e e_e + A \sum_v \left(D_v(\mathbf{e}) - 2\right)^2$$

4 customers: 10 edge bits
or 16 position bits.

A strict decoder checks every sampled tour.

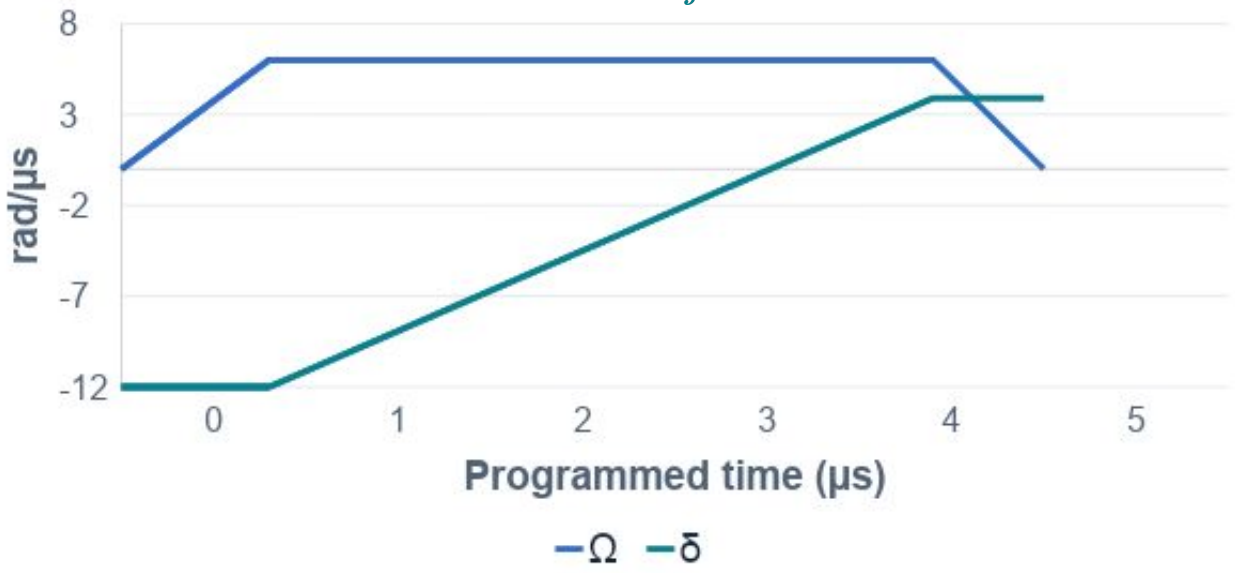
Degree = selected edges at a location.

Official benchmark: groups of ≤3 or >4 customers use classical routing. Atlas Orbit samples all three tours.

Mapping QUBO coefficients to atoms

$$\frac{H(t)}{\hbar} = \frac{\Omega(t)}{2}\sum_i X_i - \sum_i \delta_i(t) n_i + \sum_{i<j} \frac{\Gamma}{r_{ij}^6} n_i n_j$$

$$h_i^{\text{real}} = -\delta_i(T), \qquad J_{ij}^{\text{real}} = \frac{\Gamma}{r_{ij}^6}$$



Baseline pulse at 15 remaining customers / 4 vehicles

Detuning sets selection costs.

Atom separation sets pair costs.

The drive prepares the state.

5.2 µs pulse duration
Local CPU emulation; DMM adds local detuning

CPLEX leads on the 12-instance benchmark

Metric	CPLEX MILP	QUBO-assisted
Feasible full routes	36/36	33/36
Mean route gap*	0%	20.59%
Median total time	0.042 s	1.401 s
Proven optimal runs	36/36	0/36

12 instances × 3 repeats

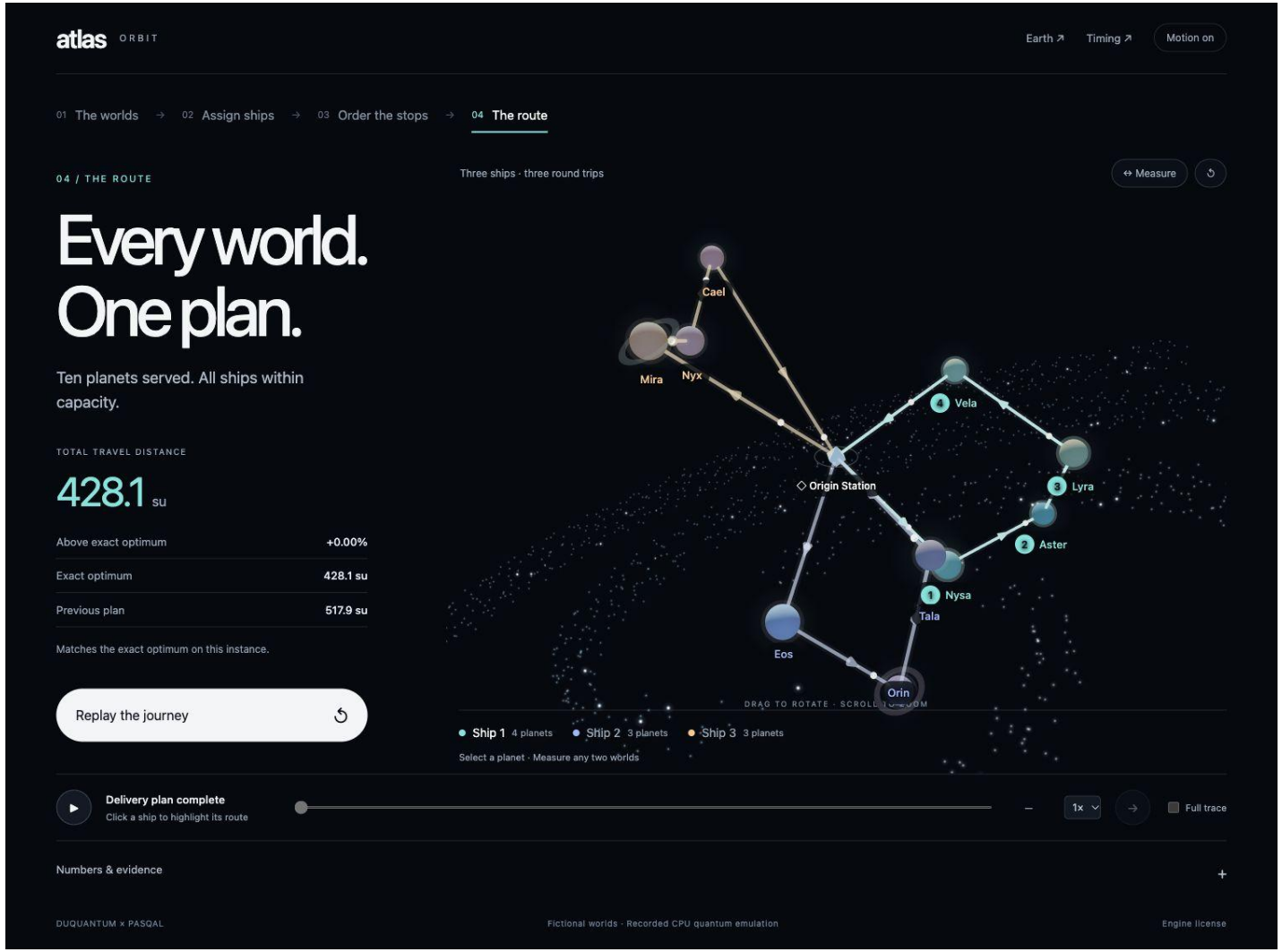
S/M/L, uniform and heterogeneous demands
Same routing objective and feasibility checks

797 / 797

target QUBO minima proven by CPLEX and checked by exhaustive enumeration

*Mean across 11 instances with successful emulator runs. CPU timings, no QPU. All 3 failures: M_heterogeneous_11.

Atlas Orbit matches the optimum on one instance



428.13 su

Matches the exact CVRP optimum

17.34% less travel

Compared with the prior 517.93 su demo run

All 3 tours sampled

10, 6 and 6 atoms
16,000 shots per ship

Fixed 3D demo. Larger budget: 512 clustering attempts, 4 window orders. CPU emulation; no speedup claim.

Where solution quality is lost

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fitted native models retain a target ground-state bitstring

Finite-time pulses can still sample useful excited states.

Clustering surrogate

Compact groups can still produce long depot tours.

Sequential capacity checks

A feasible early group can leave an infeasible remainder.

Physical embedding

Geometry distorts pair costs, even with local detuning.

What we built and what comes next

A reproducible hybrid routing experiment

QUBO derivations, measured bitstrings, strict route validation, and independent classical comparisons

Next: improve the model and embedding, then retest across more seeds

The updated Atlas Orbit demo reaches its exact optimum.

One larger-budget demo succeeds. CPLEX remains stronger on the 12-instance benchmark.

Exact assignment model and variable budget

$$E_{\text{slack}} = F(x) + A\sum_i \left(\sum_k x_{ik} - 1\right)^2 + B\sum_k \left(\sum_i d_i x_{ik} + s_k - Q\right)^2$$

$$s_k = \sum_{\ell=0}^{B_Q-1} 2^\ell u_{k\ell}, \qquad B_Q = \lceil \log_2(Q+1) \rceil$$

Encoding	S	M	L
Assignment x	8	30	60
Assignment + slack	14	45	80
Implemented window peak	4	8	8

A = B = U + 1 makes exact global minima feasible, given a feasible score U and nonnegative F.

Two routing encodings for four customers

Edge encoding: 10 binary variables

$$E_{\text{edge}} = \sum_{i<j} \tilde{c}_{ij} e_{ij} + A \sum_v \left(\sum_{e \ni v} e_e - 2 \right)^2$$

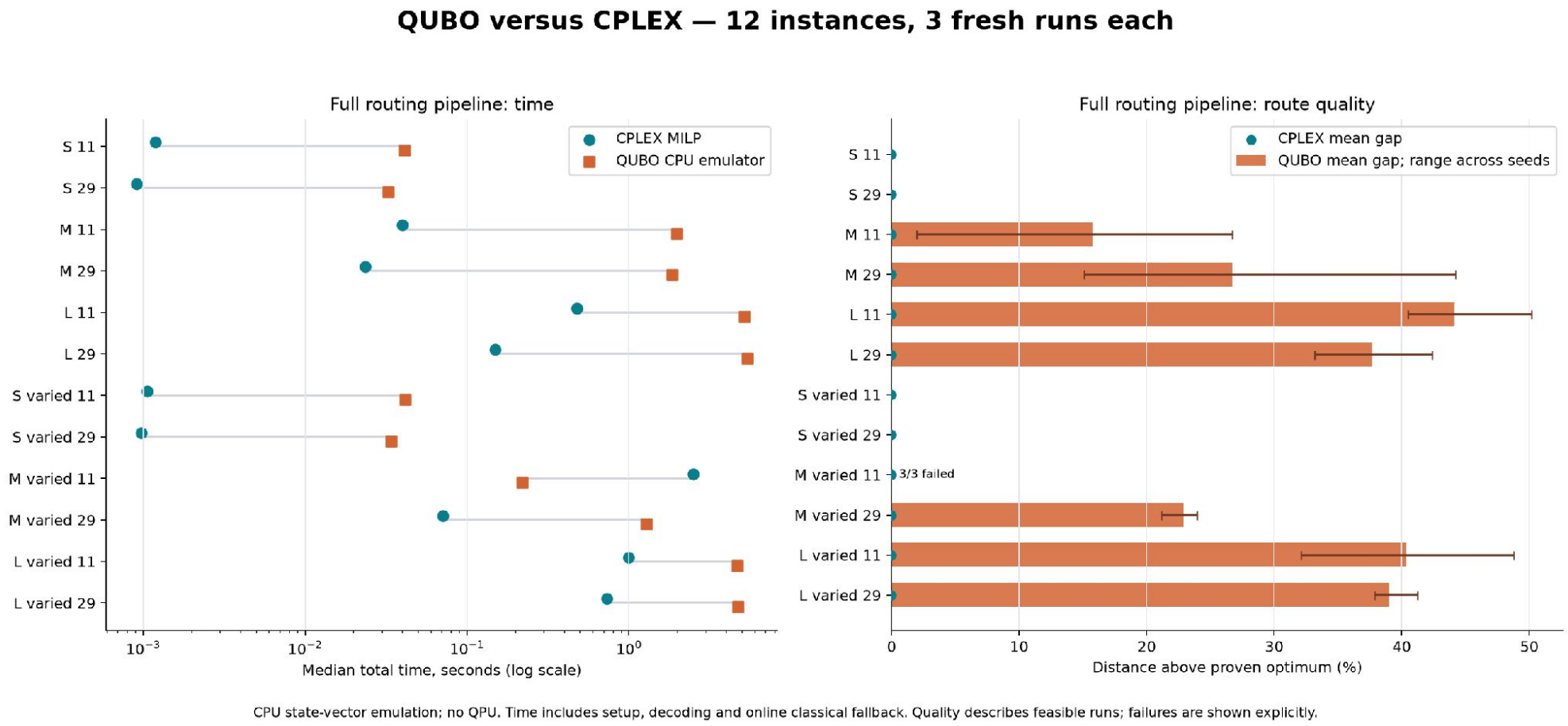
$$h_e = \tilde{c}_e - 6A, \qquad J_{ef} = 2A \quad \text{for incident edges}$$

Position encoding: 16 binary variables

$$E_{\text{pos}} = C_{\text{tour}}(x) + A \sum_i \left(\sum_t x_{it} - 1 \right)^2 + A \sum_t \left(\sum_i x_{it} - 1 \right)^2$$

Degree two implies one tour only for ≤4 customers here. Position bits encode a permutation. A = 10.

Classical comparison by instance



Local baseline runs, three repeats per instance. Failed solutions retain undefined route gaps.

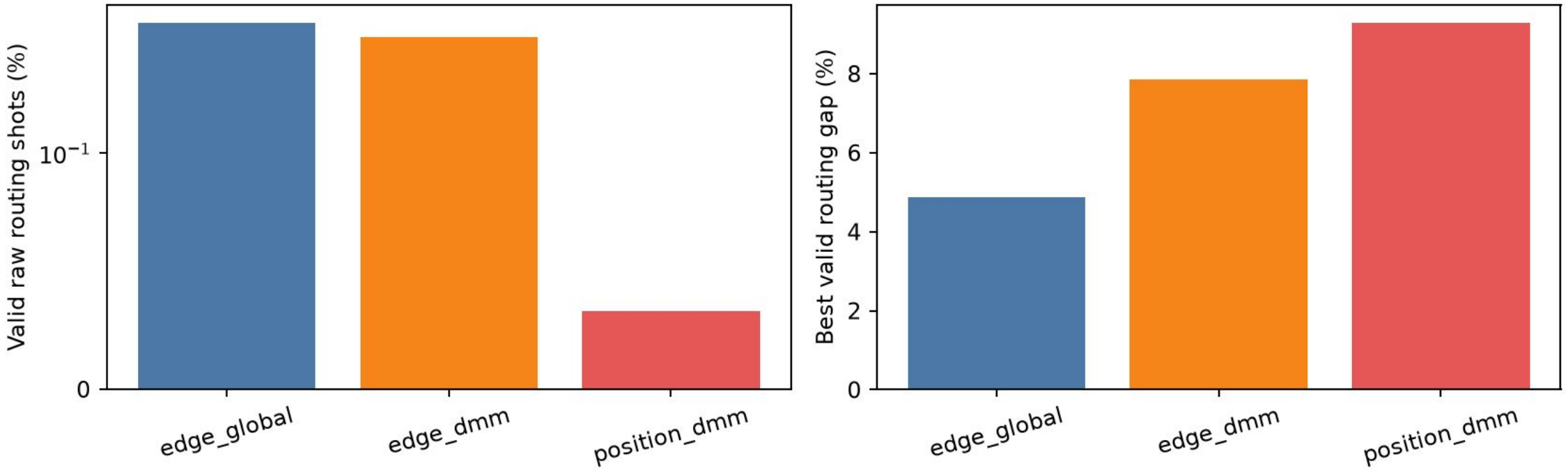
Identical-QUBO diagnostics

Stage / encoding	Banks	Sampled target optimum	Native optimum overlap
Clustering / subset	754	707/754	0/754
Routing / edge	36	19/36	0/36
Routing / position	7	0/7	0/7

Every target and native minimum passed CPLEX and exhaustive checks.

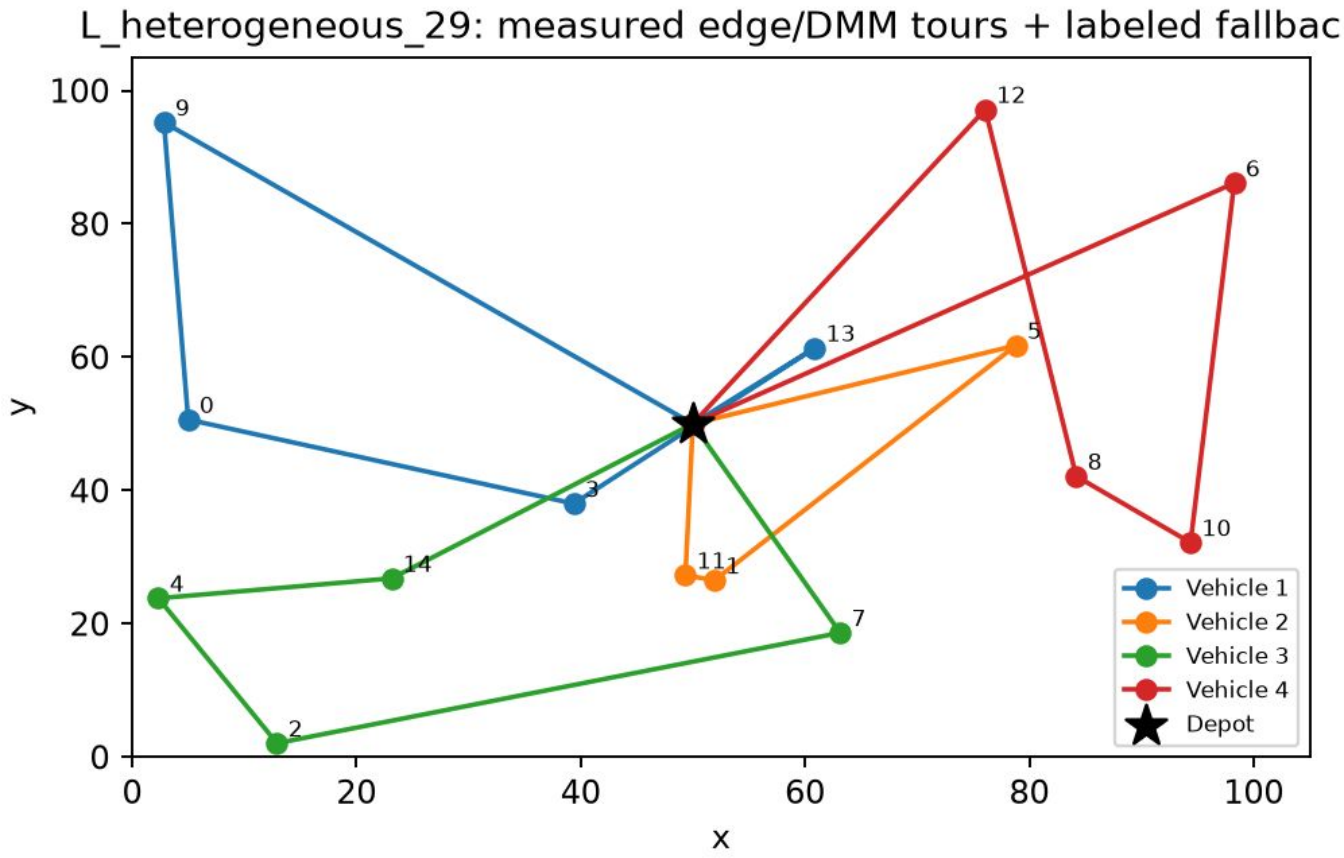
Edge banks return valid sampled tours in 36/36 cases.
Position banks return valid sampled tours in 3/7 cases.

Historical Track B encoding comparison



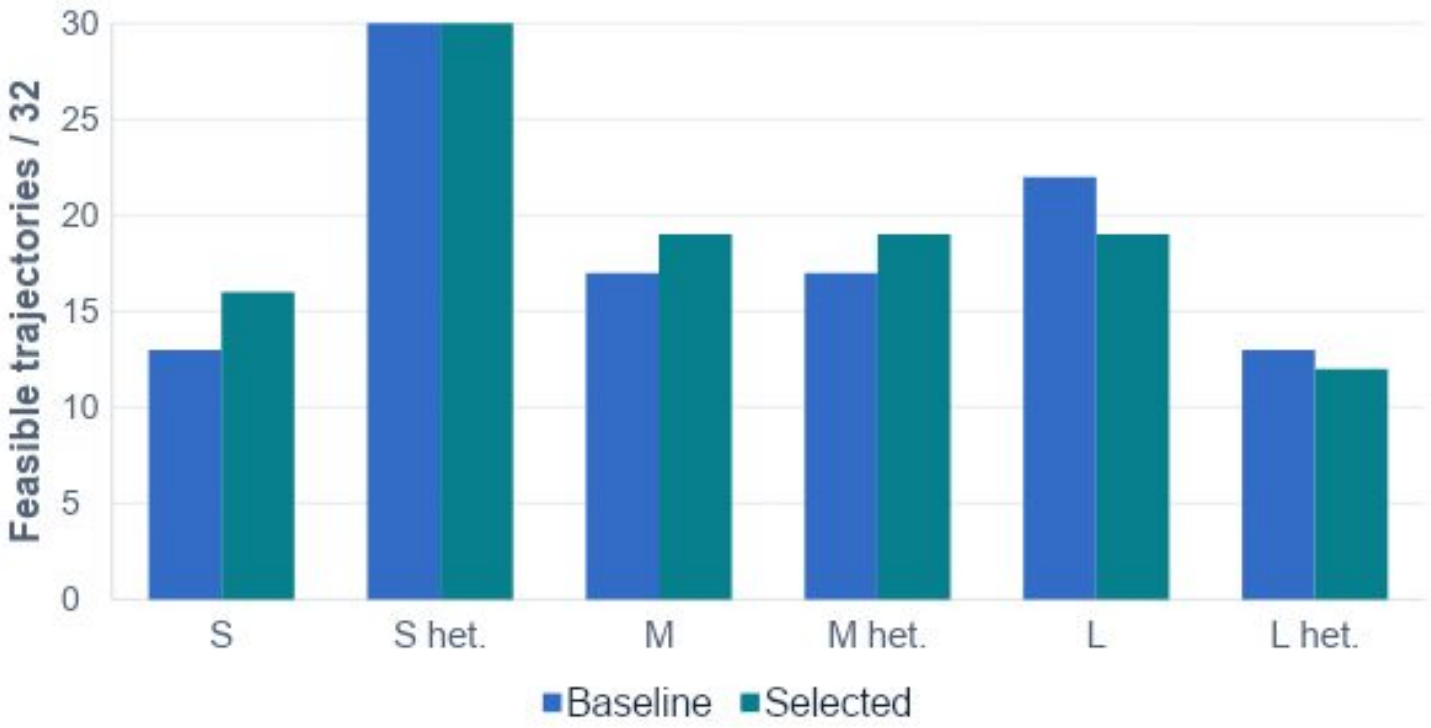
Original GitHub benchmark evidence. Different experiment from the fresh local CPLEX runs.

Historical Track B route example



Quantum routing applies to eligible four-customer groups. Trivial and size fallback groups remain classical.

Sampler tuning on six held-out cases



58.85% to 60.42%

Mean feasible trajectory fraction

8.41% to 7.57%

Mean best clustering-model gap

18.59 s to 43.42 s

Total held-out solver time

16 settings. Seed 11 training, seed 101 validation. Track A only. Separate from the latest Orbit run.

Evaluation metrics and loss accounting

Full-route optimality gap

$$\mathrm{gap}(\%) = 100\left(\frac{C}{C^*} - 1\right)$$

C* is the exact CVRP optimum. Failed runs have an undefined gap.

Where excess travel enters

$$C_{\mathrm{end}} - C_{\mathrm{ref}} = \underbrace{C_{\mathrm{model}} - C_{\mathrm{ref}}}_{\text{decomposition}} + \underbrace{C_{\mathrm{shot}} - C_{\mathrm{model}}}_{\text{clustering}} + \underbrace{C_{\mathrm{end}} - C_{\mathrm{shot}}}_{\text{routing}}$$

Absolute distances add. The clustering term can be negative because the objectives differ.